WORKSHOP PROGRAM / 26 NOVEMBER 2026

CyberNet 2026

Next-Gen Cybersecurity for Mission-Critical Mobile Networks

DATE	VENUE
Thursday, November 26, 2026	**Hotel Metropolitan Tokyo Ikebukuro — Room TBA**

1-6-1 Nishi-Ikebukuro, Toshima-ku, Tokyo, 171-8505, Japan

AT A GLANCE

TIME	SESSION
08:30–08:45	Welcome and Opening Remarks
08:45–09:45	Session 1: Intelligent and Secure Networked Systems
09:45–10:15	Coffee and Networking
10:15–11:15	Keynote 1: Prof. Wenjing Lou (Virginia Tech)
11:15–12:15	Session 2: Federated Learning and Intelligent Network Security
12:15–13:30	Lunch and Networking
13:30–14:30	Keynote 2: Prof. Mohsen Guizani (MBZUAI)
14:30–15:30	Session 3: Next-Generation Networks and Communication Security
15:30–16:00	Coffee and Networking
16:00–17:00	Session 4: Trust and Resilience in Next-Generation Networks
17:00–17:40	Session 5: Emerging Techniques in Network Intrusion Detection
17:40–17:50	Closing Remarks

TECHNICAL SESSIONS

Accepted papers & authors

08:45–09:45 | Session 1: Intelligent and Secure Networked Systems

Full-paper presentations + Q&A (20 min each)

1. Reliable V2X-Enabled Autonomous Driving with Explainable Federated Neuro-Symbolic Intelligence

Zhaoyang Du, Celimuge Wu, Yalong Li, Jiale Wu (The University of Electro-Communications); Soufiene Djahel (Coventry University); Jianhang Tang (Guizhou University)

2. RAMP: History-Aware Multi-Access Path Management for Resilient Edge Networks

Amardeep Kumar, Dinesh Sah, Lauri Loven (University of Oulu)

3. Lightweight Approximate Nearest Neighbors Based Sparse Attention Transformer for NetBIOS Attack Detection in IoT Networks

Liang Zhou (Shanghai University of Medicine and Health Sciences); Shavi Bansal (InsightsTechinfo); Akshat Gaurav (Asia University); Kwok Tai Chui (Hong Kong Metropolitan University); Mu-Yen Chen (National Cheng Kung University); Jinsong Wu (Universidad de Chile); Brij B. Gupta (Asia University)

11:15–12:15 | Session 2: Federated Learning and Intelligent Network Security

Full-paper presentations + Q&A (20 min each)

1. PQ-FL-IDS: A Post Quantum Safe Federated Edge Intrusion Detection Scheme underlying 6G

Husain Lucky, Pronaya Bhattacharya (Amity University Kolkata); Mu-Yen Chen (National Cheng Kung University); Kwok Tai Chui (Hong Kong Metropolitan University); Brij B. Gupta (Asia University); Thippa Reddy Gadekallu (Lovely Professional University)

2. Strategic Client-Level Defense Placement in Mobile Federated Learning

Anish Ambreth, Naveen Kumar Kummari, Mohsen Guizani (MBZUAI)

3. C-MoA-XRT: Explainability-Aware Trust Routing for Collaborative Threat Detection in Mobile-Edge Networks

Faisal Naeem (University of the Fraser Valley); Samuel Okegbile (Wilfrid Laurier University); Yasir Faheem (Canadian University Dubai)

14:30–15:30 | Session 3: Next-Generation Networks and Communication Security

Full-paper presentations + Q&A (20 min each)

1. Receiver-Aware Token Routing for Semantic Communication

Yangfei Lin, Celimuge Wu, Jiale Wu, Yalong Li (The University of Electro-Communications); Mohsen Guizani (MBZUAI)

2. Private 5G Security Across Public and Non-Public Networks: A Cross-Deployment Analysis

Ravishankar Borgaonkar (SINTEF Digital); Silje Kingsrud Kvisten, Thomas Zinner (NTNU)

3. A Comparative Evaluation of Modbus Security Telemetry Extraction

Erkin Kirdan, Karl Waedt (Framatome)

16:00–17:00 | Session 4: Trust and Resilience in Next-Generation Networks

Full-paper presentations + Q&A (20 min each)

1. Resilient Priority-Aware Inference for Mission-Critical UAV Networks Under Bursty Edge Load

Maher Guizani (University of Texas at Arlington); Ishita Agarwal, Naveen Kumar Kummari (MBZUAI); Mohammad Atiqul Islam (University of Texas at Arlington)

2. Trust-Driven Multi-Criteria Aerial Networking for Mission-Critical IoT with Localization Uncertainty

Muhammad Omair Butt (Lakehead University); Muhammad Naeem (COMSATS University); Waleed Ejaz (Lakehead University)

3. Latency Aware Zero Trust Authorization for Secure Mobile Edge Networks

Ankita Behera (SRM University-AP); Deepak Puthal (UAE University); Sambit Kumar Mishra (SRM University-AP); Parag Kulkarni (UAE University)

17:00–17:40 | Session 5: Emerging Techniques in Network Intrusion Detection

Full-paper presentations + Q&A (20 min each)

1. Topology-Aware Sparse Attention for Intrusion Detection in Communication Networks

Brij B. Gupta, Akshat Gaurav (Asia University); Varsha Arya (Hong Kong Metropolitan University); Mu-Yen Chen (National Cheng Kung University); Kwok Tai Chui (Hong Kong Metropolitan University)

2. Federated Transfer Learning with Attentional Graph Neural Networks for Zero-Day IoT Botnet Detection and Secure Edge Orchestration

Ajay Kumar (Bennett University); Shaktiyavesh Nandan Pratap Singh (Rajasthan Technical University, Kota); Swati Kumari (Deakin University); Ishan Budhiraja (Bennett University); Manish Chaturvedi (Rajasthan Technical University, Kota)

KEYNOTE 1 / 10:15–11:15

Featured keynote speaker



Prof. Wenjing Lou
Virginia Tech

TALK TITLE

No More Tracking, No More Robocalls: Private Communication in Public NextG Networks

ABSTRACT

Mobile networks provide ubiquitous connectivity but also expose users to privacy risks, as mobile network operators (MNOs) may be able to identify, track, and profile subscribers. These risks are particularly concerning when users rely on foreign or otherwise untrusted public networks.

In this talk, we present a novel anonymous mobile access architecture that enables legitimate subscribers to use public mobile networks without revealing their real identities or allowing their connections across different sessions to be linked. Meanwhile, the architecture preserves essential network functionality and supports user deanonymization under appropriate legal intervention. Beyond anonymous access, the architecture also supports private native mobile phone calls and a user-defined, policy-based, end-to-end mechanism for blocking unwanted communications, such as spam and scam calls. Drawing on emerging technologies such as anonymous credentials, this work explores a new approach to building private and trustworthy mobile networks for the NextG era.

BIOGRAPHY

Wenjing Lou is the W. C. English Endowed Professor of Computer Science at Virginia Tech and a Fellow of the IEEE and ACM. Her research interests cover many topics in the cybersecurity field, with her current research interest focusing on security and privacy problems in wireless networks, blockchain, trustworthy AI, and Internet of Things (IoT) systems. Prof. Lou received the Virginia Tech Alumni Award for Research Excellence in 2018, the highest university-level faculty research award. She received the INFOCOM Test-of-Time paper award in 2020 and the INFOCOM Achievement Award in 2025. She is the TPC chair for IEEE INFOCOM 2019 and ACM WiSec 2020. She was the Steering Committee Chair for IEEE CNS conference from 2013 to 2020. She is currently the steering committee chair of IEEE INFOCOM and a steering committee member of IEEE CNS. She served as a program director at the US National Science Foundation (NSF) from 2014 to 2017.

KEYNOTE 2 / 13:30–14:30

Featured keynote speaker



Prof. Mohsen Guizani

Professor and IEEE Fellow

Machine Learning Department

MBZUAI, UAE

TALK TITLE

Security & Privacy of IoT Systems with Federated Learning

ABSTRACT

To meet the Internet of Things (IoT) diverse requirements for increasing IoT smart city applications and ensure security and privacy, ML and edge computing with its privacy have recently proliferated and frequently used. An IoT edge computing device with AI/ML technologies, security related platforms and machine learning, could enable smart city systems to perform complex maneuvers that were not possible just a few years ago. So, IoT and edge computing with AI/ML have provided a platform for many of these applications to protect the data and reduce latency while increasing efficiency. These smart services/applications rely on efficient computation and communication resources. In this talk, we showcase our research activities that could contribute to improve these efforts and advocate possible solutions using such models. Then, we anticipate some challenges that need to be addressed in future research directions.

BIOGRAPHY

Mohsen Guizani (Fellow, IEEE) received the BS (with distinction), MS and PhD degrees in Electrical and Computer engineering from Syracuse University, Syracuse, NY, USA. He is currently a Professor of Machine Learning at the Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE. Previously, he worked in different institutions in the USA. His research interests include applied machine learning and artificial intelligence, smart city, Internet of Things (IoT), intelligent autonomous systems, and cybersecurity. He became an IEEE Fellow in 2009 and was listed as a Clarivate Analytics Highly Cited Researcher in Computer Science, 2019 to 2022. Dr. Guizani has won several research awards including the “2015 IEEE Communications Society Best Survey Paper Award”, the Best ComSoc Journal Paper Award in 2021 as well 5 Best Paper Awards from ICC and Globecom Conferences. He is the author of 11 books, more than 1000 publications and several US patents. He is also the recipient of the 2017 IEEE Communications Society Wireless Technical Committee (WTC) Recognition Award, the 2018 AdHoc Technical Committee Recognition Award, and the 2019 IEEE Communications and Information Security Technical Recognition (CISTC) Award. He served as the Editor-in-Chief of IEEE Network and is currently serving on the Editorial Boards of many IEEE Transactions and Magazines. He was the Chair of the IEEE Communications Society Wireless Technical Committee and the Chair of the TAOS Technical Committee. He served as the IEEE Computer Society Distinguished Speaker and is currently the IEEE ComSoc Distinguished Lecturer.